MODULE 14 · DOMAIN 3 - RESILIENT CLOUD SOLUTIONS

High Availability and Multi-Region Design

Surviving the loss of an Availability Zone or a Region: data replication, DNS and health checks, the edge, and Application Recovery Controller.

DOMAIN 3 of 6 EXAM WEIGHT 15% MODULE 14 of 34

admin@sthithapragna.com

Resilience is three separate problems: **the data** must already be there, **the traffic** must find it, and **someone or something** must decide to move.

Zone failure or Region failure?

FAILURE	TYPICAL PROTECTION	COST
An instance	Auto Scaling replaces it	Almost nothing
An Availability Zone	Multi-AZ everything: subnets, load balancer, database	Modest
A Region	A second Region with data replicated and traffic steerable	Significant
A bad deployment	Rollback, and shifting traffic away from where it landed	Process, not infrastructure

ON THE EXAM

Multi-AZ is the default expectation. Multi-Region needs a stated requirement: a Regional RTO, or regulation.

Multi-AZ, done properly

- Subnets and instances in at least two, ideally three, Availability Zones behind a load balancer.
- **RDS Multi-AZ**: one standby for failover, or a Multi-AZ **cluster** with two readable standbys.
- Enough capacity that losing one zone still leaves enough to serve: static stability.
- A NAT gateway per zone. One shared NAT gateway is a single-zone dependency.

> ON THE EXAM
>
> Statically stable: survive a zone loss without launching anything new, because launching may fail during the event.

Getting data to another Region

DATA STORE	CROSS-REGION OPTION	WRITES IN
Aurora	Global Database: storage-level replication, typically under a second	One Region
DynamoDB	Global tables: eventual, or strongly consistent	Every Region
S3	Cross-Region Replication, with RTC for a 15-minute objective	Each bucket
RDS (non-Aurora)	Cross-Region read replica, promoted in a disaster	One Region
ElastiCache	Global Datastore	One Region

ON THE EXAM

Only DynamoDB global tables accept writes in every Region. Everything else has one writer and needs a promotion.

Aurora Global Database: switchover or failover?

SWITCHOVER

Planned, no data loss

- Waits for the secondary to fully catch up
- Swaps writer and reader Regions
- Global topology kept intact
- For DR tests and Regional rotation

FAILOVER

Unplanned, with allow-data-loss

- Promotes a secondary during an outage
- RPO of seconds: recent writes may be lost
- Topology rebuilt once the old Region returns
- For real disasters only

THE TIEBREAKER

Planned move with no data loss: switchover. Region down: failover. Detaching and promoting breaks the global database.

DynamoDB global tables: eventual or strong?

STRONG CONSISTENCY (MRSC)

Zero RPO

- Writes replicated before they succeed
- Strongly consistent reads in any Region
- Exactly three Regions, or two plus a witness
- Higher write latency; no transactions

EVENTUAL CONSISTENCY (MREC)

The default

- Asynchronous, usually within a second
- Last writer wins on conflicts
- Any number of Regions
- RPO greater than zero

THE TIEBREAKER

Generally available June 2025. The mode is chosen at creation and cannot be changed later.

S3 across Regions

- **Replication** needs versioning on both buckets and copies new objects only.
- **Batch Replication** copies objects that existed before the rule, or failed earlier.
- **Replication Time Control**: 99.99% of objects within 15 minutes, with metrics and events.
- **Multi-Region Access Points** give one global endpoint, active-active or active-passive with failover controls.

> ON THE EXAM
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> 'Replicated within 15 minutes, guaranteed': RTC. 'One endpoint, fail over between buckets': a Multi-Region Access Point.

Route 53 routing policies

POLICY	SENDS TRAFFIC TO
Failover	Primary while healthy; secondary when not
Latency	The Region with lowest latency for the user
Weighted	Endpoints in proportion to weights: 90/10, for example
Geolocation	Endpoints by the user's country or continent
Geoproximity	Endpoints by distance, adjustable with a bias
Multivalue answer	Up to eight healthy records, chosen at random

ON THE EXAM

Health checks make every policy smarter: unhealthy endpoints are left out of answers.

Health checks that tell the truth

- **Endpoint** checks probe a URL or port from locations around the world.
- **Calculated** checks combine others: healthy if at least two of three are.
- **CloudWatch alarm** checks follow an alarm, which is how you watch private resources.
- **Evaluate target health** on alias records lets Route 53 use a load balancer's own health.

> ON THE EXAM
>
> Check a deep health page that tests dependencies, not just the web server. A shallow check fails over too late.

Route 53 or Global Accelerator?

GLOBAL ACCELERATOR

Anycast IPs on the AWS network

- Two static IP addresses, worldwide
- Failover in seconds, no DNS caching
- TCP and UDP
- Traffic dials and endpoint weights

ROUTE 53

DNS answers

- Changes the answer clients receive
- Failover bounded by TTL and client caching
- Any protocol, since it is only DNS
- Cheaper and simpler

THE TIEBREAKER

Static IPs, UDP, or failover faster than DNS caching allows: Global Accelerator. Otherwise Route 53 is usually enough.

Resilience at the edge: CloudFront

- An **origin group** pairs a primary and secondary origin, such as S3 buckets in two Regions.
- On chosen error codes or timeouts, CloudFront retries the request on the secondary.
- Origin failover applies to **GET, HEAD and OPTIONS**. Writes are not failed over.
- **Origin Shield** adds a central cache layer, reducing load on the origin.

> ON THE EXAM
>
> Static content that must survive a Regional S3 failure: an origin group across two replicated buckets.

Application Recovery Controller

CAPABILITY	WHAT IT DOES
Zonal shift	Moves traffic away from one AZ, for up to three days, then expires
Zonal autoshift	AWS starts the shift itself when it detects a zone problem
Routing control	Highly available on/off switches for Route 53 health checks
Region switch (August 2025)	Orchestrated, multi-step, cross-account Region failover plans
Readiness check	Closed to new customers from 30 April 2026

ON THE EXAM

Zonal shift and autoshift are free. Region switch plans evaluate themselves every 30 minutes.

Why recovery tools must be independent

- Control planes, which create and change resources, are less available than data planes.
- Recovery should use **data-plane** actions: health checks, routing controls, already-running capacity.
- A failover that needs a console, or a resource launched in the failing Region, can fail with it.
- Region switch plans are replicated to every Region they cover, so no Region depends on another.

> ON THE EXAM
>
> Rehearse failover regularly. An untested recovery plan is a hope, not a plan.

Active-active or active-passive?

ACTIVE-ACTIVE

Every Region serves traffic

- Failover is just removing a Region
- Capacity already warm everywhere
- Data must accept writes in each Region
- Conflicts must be designed for

ACTIVE-PASSIVE

One Region serves; one waits

- Simpler data: one writer
- Standby may need scaling up first
- Failover needs a promotion and a switch
- Cheaper, if the standby is small

THE TIEBREAKER

Active-active needs a multi-writer data layer, such as DynamoDB global tables. With Aurora, the writer is always in one Region.

Load balancers during failures and deployments

- Target group **health checks** remove unhealthy targets; set path, interval and thresholds deliberately.
- **Deregistration delay** lets in-flight requests finish before a target is removed. Default 300 seconds.
- **Slow start** ramps traffic to a new target gradually, for applications that warm up.
- **Cross-zone load balancing** spreads traffic evenly across targets in all zones. Always on for ALB.

> ON THE EXAM
>
> Requests fail as instances are removed during scale-in or deployments: check the deregistration delay.

> A payments company stores account balances in DynamoDB. It runs in several Regions so that it can survive a Regional outage.
>
> After a Regional failure, no acknowledged write may be lost, and any Region must return the latest balance on a strongly consistent read. Latency on writes is acceptable.

What should the company use?

A. A DynamoDB global table with multi-Region strong consistency, in three Regions or two plus a witness

B. A DynamoDB global table with the default eventual consistency mode

C. DynamoDB Streams with a Lambda function writing to a table in another Region

D. Point-in-time recovery, with backups copied to another Region

◆ PAUSE AND DECIDE

What should the company use?

A Global table, strong consistency

Writes replicate before succeeding: zero RPO, and strongly consistent reads in any Region.

B Global table, eventual

Asynchronous: the last second of writes can be lost, and remote reads can be stale.

C Streams and Lambda

Custom, asynchronous replication. Worse than B on every requirement.

D Backups copied across

Restores lose everything since the last backup, and take time.

THE SIGNAL

Zero RPO plus strongly consistent reads anywhere: MRSC. 'Write latency is acceptable' is the planted permission.

> A multiplayer game uses UDP, with game servers behind Network Load Balancers in two Regions. Some players' networks allow outbound traffic only to IP addresses on an approved list.
>
> When a Region fails, players must reach the other Region within seconds. Past Route 53 failovers took many minutes for some players because of DNS caching.

Which solution meets these requirements?

A. Route 53 failover routing with a 10-second TTL

B. AWS Global Accelerator with an endpoint group in each Region and health checks

C. CloudFront with an origin group containing both load balancers

D. Cross-zone load balancing enabled on both Network Load Balancers

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A Shorter TTL

Clients that ignore TTL still cache. And IP addresses still change, breaking allow-lists.

B Global Accelerator

Two static IPs for allow-lists, UDP support, and failover in seconds with no DNS involved.

C CloudFront origin group

CloudFront serves HTTP and HTTPS. It does not carry game UDP traffic.

D Cross-zone

Spreads load across zones in one Region. Nothing to do with Regional failover.

THE SIGNAL

Static IPs, UDP, and failover that DNS caching cannot slow: every clue points to Global Accelerator.

> A company's Aurora Global Database has its writer in us-east-1 and a secondary in us-west-2. Both Regions are healthy.
>
> For an annual disaster recovery test, the company must move the writer to us-west-2 with no data loss, and keep the global database intact so it can move back afterwards.

What should the DevOps engineer do?

A. Perform a managed switchover of the global database to us-west-2

B. Perform a failover to us-west-2 using the allow-data-loss option

C. Detach the us-west-2 cluster from the global database and promote it

D. Copy a snapshot to us-west-2 and restore it as a new cluster

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Switchover

Waits for full synchronisation, swaps Regions, keeps the topology. Built for exactly this.

B Failover with data loss

For outages. Accepts losing recent writes, which this test must not.

C Detach and promote

Breaks the global database, so there is nothing to move back with.

D Snapshot and restore

Loses writes after the snapshot, takes far longer, and creates an unrelated cluster.

THE SIGNAL

Both Regions healthy, planned, no data loss, move back later: switchover. Failover is for when a Region is down.

> A web application runs behind an Application Load Balancer across three Availability Zones. A faulty deployment reached only the instances in one zone, and requests there are failing.
>
> The team wants to move traffic away from that zone within minutes, reverse it easily later, and have AWS do the same automatically if it detects a zone problem in future.

What should the team use?

A Deregister that zone's instances from the target group by hand

B ARC zonal shift now, and enable zonal autoshift for the load balancer

C Route 53 failover routing to a second Region

D Remove that zone's subnet from the load balancer

◆ PAUSE AND DECIDE

What should the team use?

A. Deregister by hand

Works slowly, must be undone by hand, and nothing is automatic next time.

B. Zonal shift and autoshift

Moves traffic away in one action, expires or is cancelled to reverse, and autoshift covers the future.

C. Fail over to another Region

Two healthy zones remain. A Regional failover is far out of proportion.

D. Remove the subnet

A configuration change to the load balancer, harder to reverse, and not automatic.

THE SIGNAL

One zone bad, two good: shift away from the zone, not the Region. Autoshift makes AWS do it for you next time.

> An application in two Regions uses Route 53 failover routing. The primary Region's health really depends on an internal database that is only reachable inside its VPC.
>
> The web tier sometimes stays up while the database is failing, so the endpoint health check passes and no failover happens.

How should the DevOps engineer make failover reflect the database's health?

A. Point a Route 53 endpoint health check at the database's private IP address

B. Replace the check with a calculated health check containing only the web tier check

C. Create a CloudWatch alarm on the database's health metrics, and a Route 53 health check that follows the alarm

D. Write a Lambda function that edits the DNS record when it detects a database problem

◆ PAUSE AND DECIDE

How should the DevOps engineer make failover reflect the database's health?

A **Endpoint check on a private IP**

Route 53 checkers run on the internet and cannot reach private addresses.

B **Calculated check of the web tier**

Combines the same shallow check that already misses the problem.

C **Alarm-based health check**

The alarm sees private metrics; the health check follows it; failover then reflects the database.

D **Lambda edits DNS**

Custom failover logic running in the Region that is failing. Rebuilds what alarm checks do.

THE SIGNAL

Private resource, Route 53 failover: a health check driven by a CloudWatch alarm.

> An Auto Scaling group runs 4 instances across two Availability Zones, sized for normal peak load. During a zone outage, the group tried to launch replacements in the surviving zone, but launches were slow and the service degraded for an hour.
>
> The company wants the application to survive the loss of any one zone without depending on launching new instances during the event.

What should the DevOps engineer do?

A. Add a predictive scaling policy to the group

B. Replace the instances with larger instance types

C. Add a warm pool of stopped instances in the same two zones

D. Spread across three zones, with enough running capacity that any two zones can carry peak load

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A **Predictive scaling**

Forecasts demand. It still launches instances, which is what failed.

B **Larger instances**

Losing one of two zones still halves capacity, however large each instance is.

C **Warm pool**

Faster launches, but still launches during the event, and one zone of two is still half.

D **Three zones, N+1 capacity**

Losing a zone leaves enough running capacity. Nothing needs launching: static stability.

THE SIGNAL

'Without launching anything during the event' is static stability: pre-provisioned capacity that survives a zone loss.

Module 14 in five lines

- Three problems: data already there, traffic able to find it, and something that decides to move.
- Only DynamoDB global tables write everywhere; MRSC gives zero RPO across exactly three Regions.
- Aurora: switchover for planned moves, failover with data loss for outages.
- Route 53 is bounded by DNS caching; Global Accelerator gives static IPs and fast failover.
- ARC: zonal shift and autoshift for zones; Region switch for orchestrated Regional failover.

> ON THE EXAM
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> Next, module 15: scaling everything. Auto Scaling, ECS, EKS, Lambda, databases and queues.